

Assignment 6

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1. Find the analytical expressions for the magnitude and phase response for each $G(s)$ below.

A.

$$G(s) = \frac{1}{s(s+2)(s+4)}, H(s) = 1$$

$$|G(s)| = \frac{1}{|jw| * |jw+2| * |jw+4|}$$

Magnitude:

$$20\log_{10}|G(s)| = 20\log_{10}\left(w^{-1} * (w^2+4)^{-\frac{1}{2}} * (w^2+16)^{-\frac{1}{2}}\right)$$

$$20\log_{10}|G(s)| = -20\log_{10}(w) - 10\log_{10}(w^2+4) - 10\log_{10}(w^2+16)$$

Phase:

$$\angle G(jw) = -\left(\angle(jw) + \angle(jw+2) + \angle(jw+4)\right)$$

$$\angle G(jw) = -\left(90^\circ + \tan^{-1}\left(\frac{w}{2}\right) + \tan^{-1}\left(\frac{w}{4}\right)\right)$$

B.

$$G(s) = \frac{s+5}{s(s+2)(s+4)}, H(s) = 1$$

$$|G(s)| = \frac{|jw+5|}{jw * |jw+2| * |jw+4|}$$

Magnitude:

$$20\log_{10}|G(s)| = \left(10\log_{10}(w^2+25) - 20\log_{10}(w^2) - 20\log_{10}(w^2+4) - 10\log_{10}(w^2+16)\right)$$

Phase:

$$\angle G(s) = \tan^{-1}\left(\frac{w}{5}\right) - \left(90^\circ + \tan^{-1}\left(\frac{w}{2}\right) + \tan^{-1}\left(\frac{w}{4}\right)\right)$$

C.

$$G(s) = \frac{(s+5)(s+3)}{s(s+2)(s+4)}, H(s) = 1$$

Magnitude:

$$|G(s)| = \frac{|jw+5| * |jw+3|}{|jw| * |jw+2| * |jw+4|}$$

$$20\log_{10}|G(s)| = 10\log_{10}(w^2 + 25) + 10\log_{10}(w^2 + 9) - \left(20\log_{10}(w^2) + 10\log_{10}(w^2 + 4) + 10\log_{10}(w^2 + 16) \right)$$

Phase:

$$\angle G(s) = \tan^{-1}\left(\frac{w}{5}\right) + \tan^{-1}\left(\frac{w}{3}\right) - \left(90^\circ + \tan^{-1}\left(\frac{w}{2}\right) + \tan^{-1}\left(\frac{w}{4}\right) \right)$$

2. Find the Bode diagram (by hand) for each of the following systems.